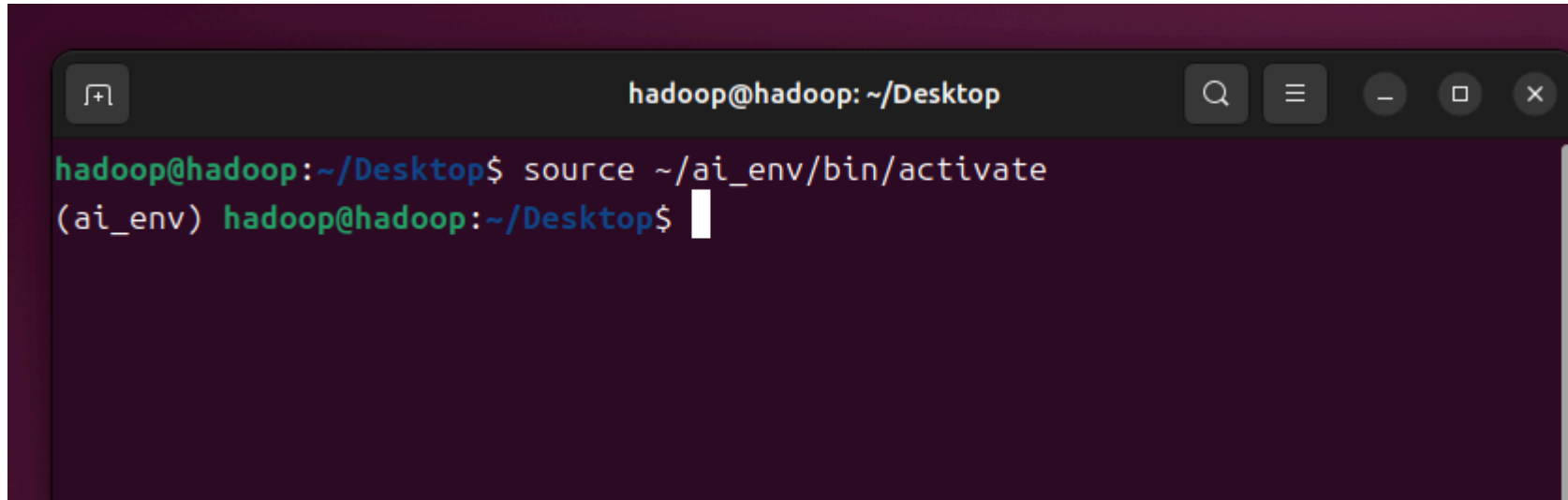
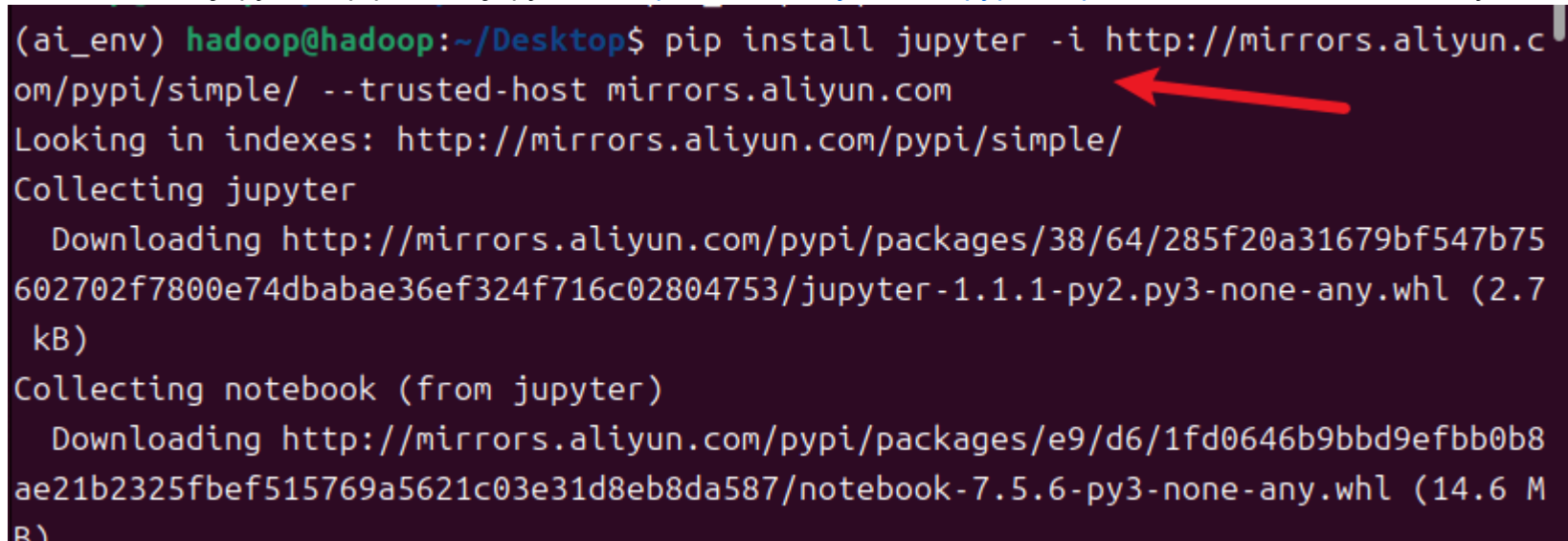


在实训的过程当中，我们有可能需要用到jupyter notebook，这就要求安装jupyter模块。1.打开终端，激活虚拟环境（或者直接使用pycharm当中的终端，可自动激活） `source ~/ai_env/bin/activate`

A terminal window with a dark purple background. The title bar shows 'hadoop@hadoop: ~/Desktop'. The prompt is 'hadoop@hadoop:~/Desktop\$'. The command 'source ~/ai_env/bin/activate' has been entered, and the prompt has changed to '(ai_env) hadoop@hadoop:~/Desktop\$' with a white cursor at the end.

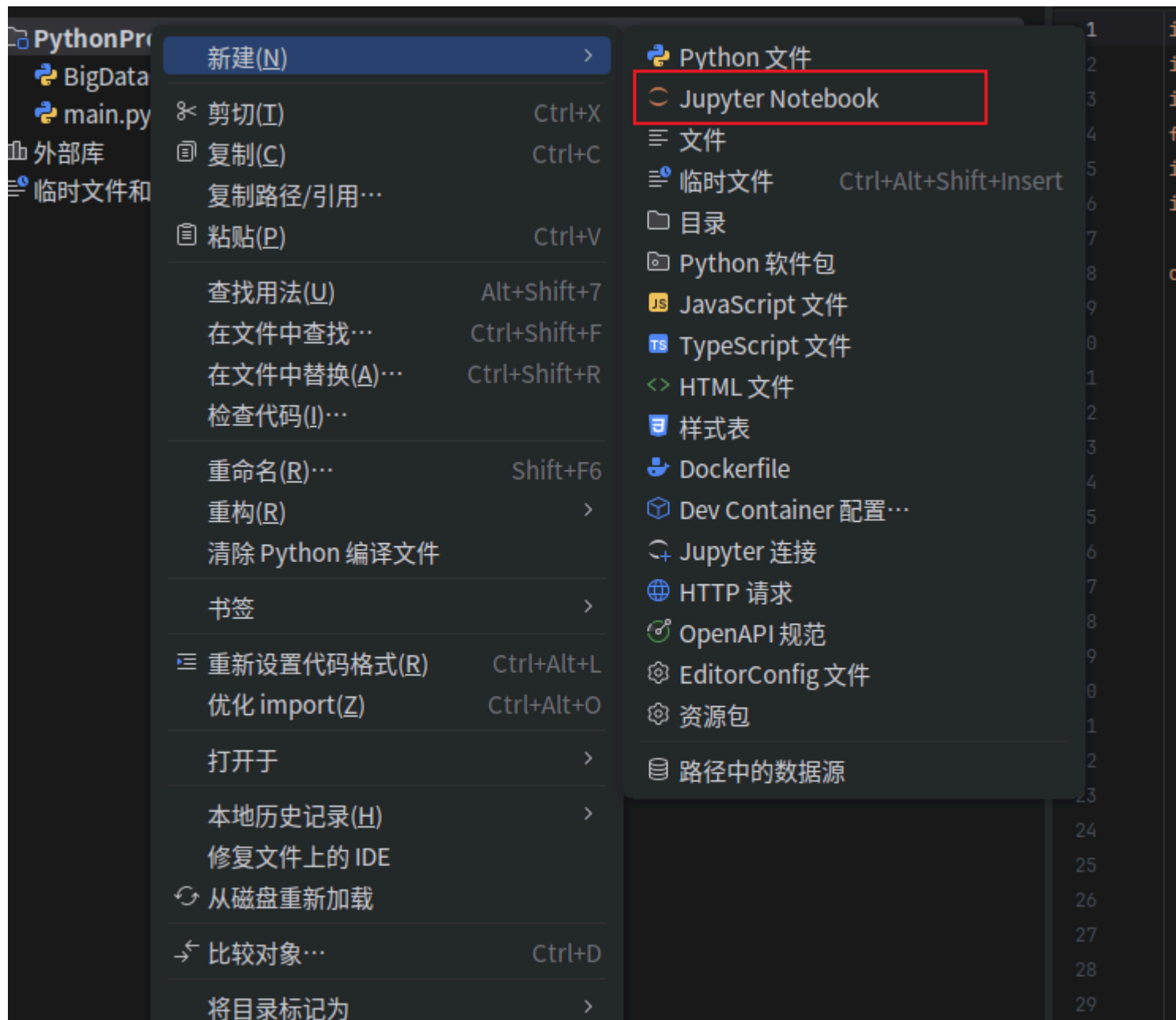
```
hadoop@hadoop:~/Desktop$ source ~/ai_env/bin/activate
(ai_env) hadoop@hadoop:~/Desktop$
```

2.输入指令安装jupyter库 `pip install jupyter -i http://mirrors.aliyun.com/pypi/simple/ --trusted-host mirrors.aliyun.com`

A terminal window with a dark purple background. The prompt is '(ai_env) hadoop@hadoop:~/Desktop\$'. The command 'pip install jupyter -i http://mirrors.aliyun.com/pypi/simple/ --trusted-host mirrors.aliyun.com' has been entered. A red arrow points to the URL 'http://mirrors.aliyun.com/pypi/simple/'. The output shows the package being collected and downloaded from the mirror, followed by the collection and download of the notebook package.

```
(ai_env) hadoop@hadoop:~/Desktop$ pip install jupyter -i http://mirrors.aliyun.com/pypi/simple/ --trusted-host mirrors.aliyun.com
Looking in indexes: http://mirrors.aliyun.com/pypi/simple/
Collecting jupyter
  Downloading http://mirrors.aliyun.com/pypi/packages/38/64/285f20a31679bf547b75602702f7800e74dbabae36ef324f716c02804753/jupyter-1.1.1-py2.py3-none-any.whl (2.7 kB)
Collecting notebook (from jupyter)
  Downloading http://mirrors.aliyun.com/pypi/packages/e9/d6/1fd0646b9bbd9efbb0b8ae21b2325fbef515769a5621c03e31d8eb8da587/notebook-7.5.6-py3-none-any.whl (14.6 MB)
```

3.打开pycharm, 在项目当中新建一个Jupyter Notebook



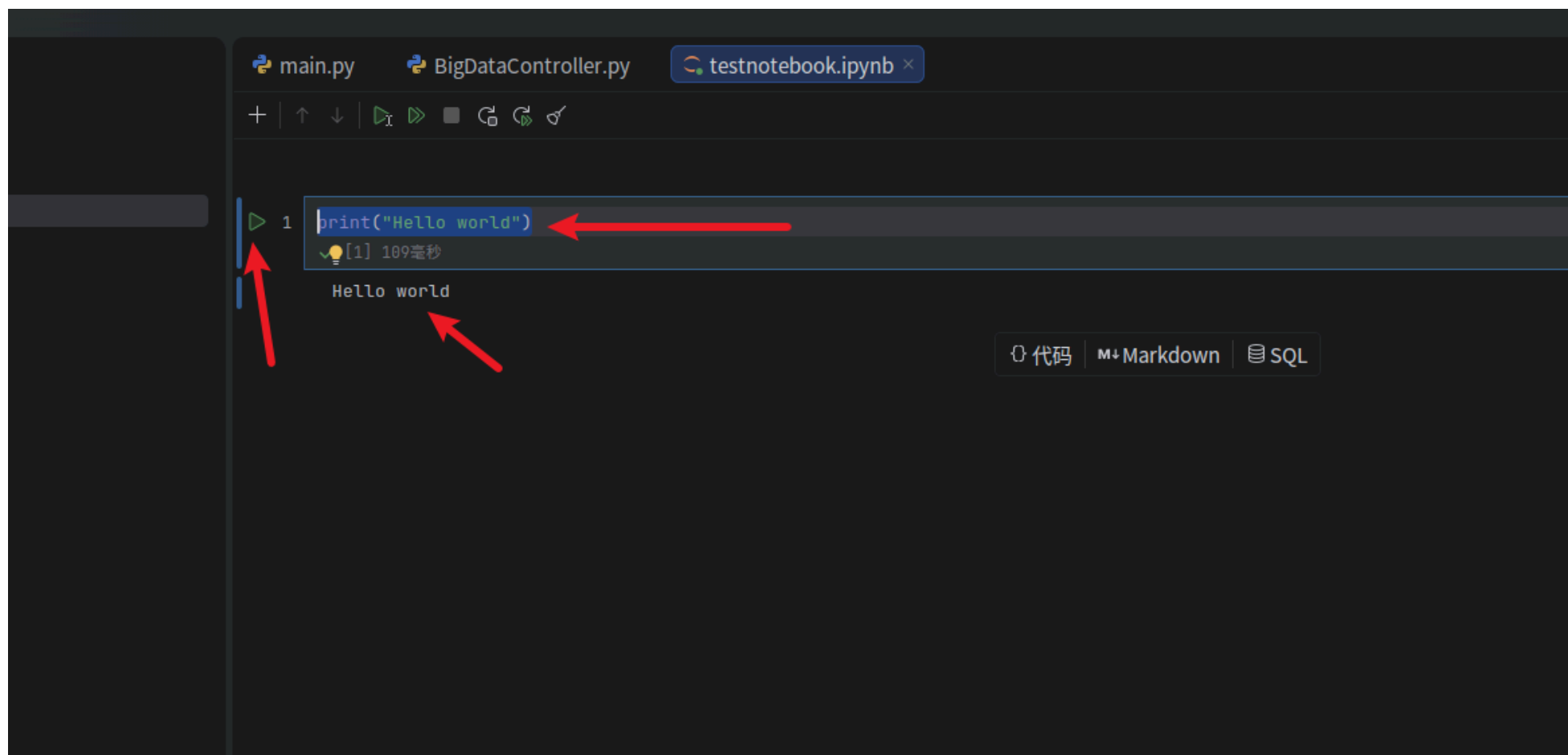
图表

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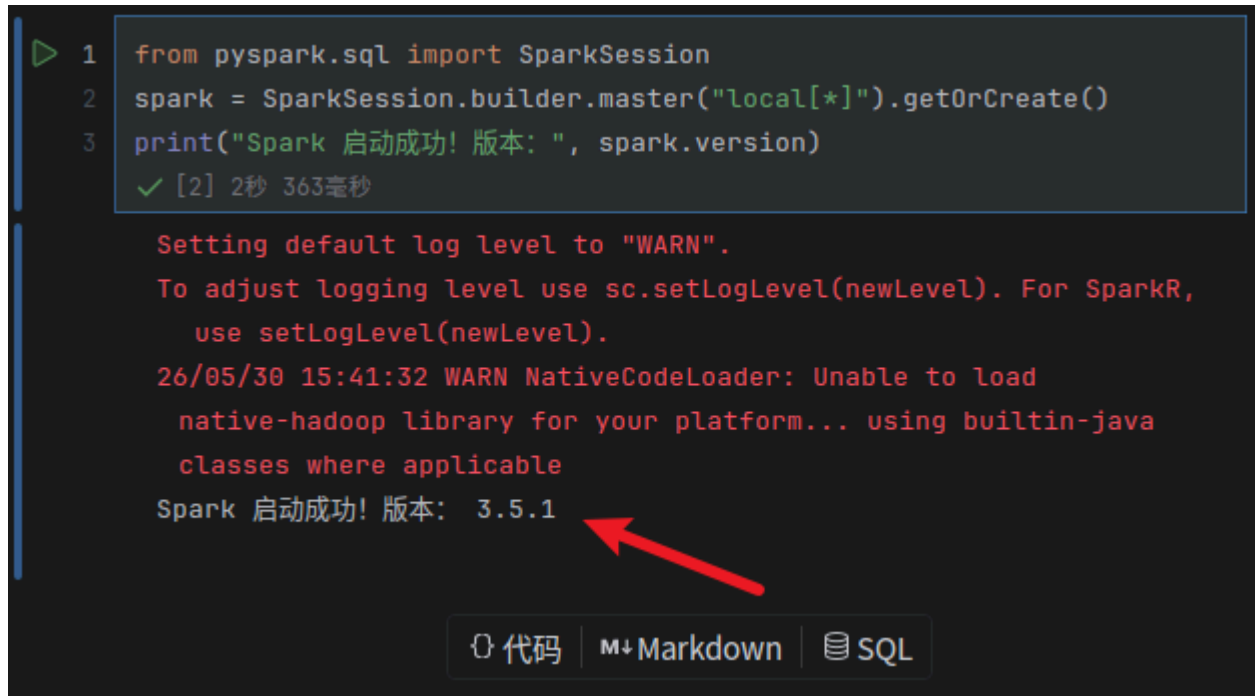
4.在代码块中输入Python代码，点击左侧运行按钮运行，如果能看到运行结果，证明Jupyter安装成功。

```
print("Hello world")
```



5.新建一个代码块，输入如下代码。如果运行成功，会显示spark版本号。安装完毕，请关闭虚拟机并拍摄快照

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.master("local[*]").getOrCreate()
print("Spark 启动成功! 版本: ", spark.version)
```



```
1 from pyspark.sql import SparkSession
2 spark = SparkSession.builder.master("local[*]").getOrCreate()
3 print("Spark 启动成功! 版本: ", spark.version)
✓ [2] 2秒 363毫秒
```

Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR,
use setLogLevel(newLevel).
26/05/30 15:41:32 WARN NativeCodeLoader: Unable to load
native-hadoop library for your platform... using builtin-java
classes where applicable
Spark 启动成功! 版本: 3.5.1

代码 | Markdown | SQL